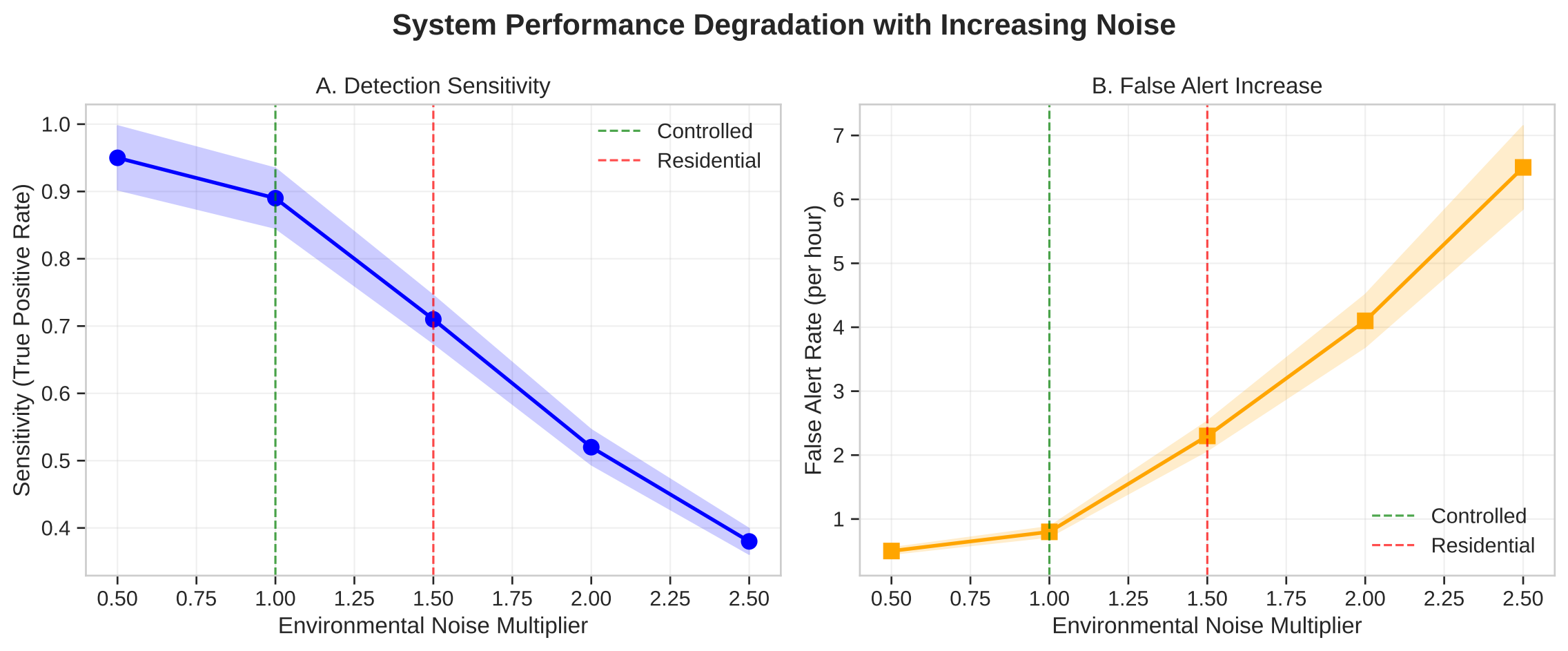
System Performance Degradation with Increasing Noise
A. Detection Sensitivity
Sensitivity (True Positive Rate)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
Controlled
Residential
0.50
0.75
1.00
1.25
1.50
1.75
2.00
2.25
2.50
Environmental Noise Multiplier
B. False Alert Increase
False Alert Rate (per hour)
7
6
5
4
3
2
1
Controlled
Residential
0.50
0.75
1.00
1.25
1.50
1.75
2.00
2.25
2.50
Environmental Noise Multiplier